

# **A review of the transport sector: Notes from the post-COVID-19 era**

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## **Abstract**

The transport industry is one of the fundamental development pillars of all socioeconomic systems nowadays. This article aims to examine whether the current pandemic crisis of COVID-19—which significantly repositions all socioeconomic dimensions—accelerates developments, especially in the transport sector. This introduction investigates critical perspectives from the history of transport, ending up with the trends that (seemingly) existed before the current crisis. The primary finding corresponds to the acceleration caused by COVID-19, especially in trends related to increased digitalization and the need for investment in new infrastructure across the world. Also, the de-carbonization and the rise of the “prosumer” (the individual who consumes and produces value simultaneously) are accelerating forces in the transport industry.

**Keywords:** transport sector, COVID-19 crisis, history of transport, investment infrastructure, digitalization, de-carbonization, prosumer

## **1. Introduction**

The transport industry is one of the fundamental development pillars of today’s global socioeconomic system. This study aims to present related trends in the transport system, in conjunction with developments occurring after the outbreak of the current pandemic crisis of COVID-19. Various studies in the literature examine the evolution of the transport sector (Bradford, 2018; Zeller, 2014). However, by researching tendencies accelerated by COVID-19 and exploring their relation to the transport industry and the fourth industrial revolution valuable aspects of tomorrow’s global socioeconomic development can be illuminated.

This conceptual article presents the transport sector from a historical perspective by elliptically exploring the related literature (Jaakkola, 2020; Yadav, 2010). It aims to end up with a repositioned theoretical framework capable of highlighting trends and future research directions in transport for the post-COVID-19 era. Section two reviews critical elements from this sector’s historical formation. Section three investigates whether COVID-19 accelerates the existing tendencies. Section four concludes and discusses the prospects for further research.

## **2. The transport industry from a historical perspective and present trends**

The history of transport is an interdisciplinary scientific field, investigating the genesis and evolution of the sector and the main modes of public transport in various national or local settings (Gorbunov et al., 2019). The long-term evolution of the transport sector involves successive replacements in terms of technology, where new transport modes (in terms of better quality, efficiency, and speed) substitute the traditional forms (Grubler & Nakicenovic, 1991).

According to the neo-Schumpeterian perspective of the techno-economic paradigm, the entire capitalist system<sup>1</sup> follows a cyclical behavior, where long “Kondratiev waves” lead to generalized depressions every fifty to sixty years (Chatzinikolaou & Vlados, 2019b; Kondratieff, 1935; Perez, 1983). The diffusion of sufficient institutional and social innovations in the new wave drives rapid growth and investment infrastructure, paving the way for the assimilation of the techno-economic paradigm (Chatzinikolaou & Vlados, 2019a). In this context, the global system traverses nowadays the fifth Kondratiev wave, based on information technology (from 1990 and beyond). This last wave succeeded the first (1800—steam engine and cotton), the second (1850—railway and steel), the third (1900—electrical engineering and chemistry), and the fourth (1950—petrochemicals and automobiles). Others suggest that today’s world economy enters with increased velocity in the sixth wave, induced by transformations occurring in additive, nano, cognitive, information, and medical technologies (Grinin & Korotayev, 2020). Also, this sixth Kondratiev wave shares elements with the fourth industrial revolution theory, whose primary conceptual platform is the ongoing spread of “cyberphysical” systems—the outcome of the blurring boundaries between the physical and digital world in the production of products and services (Schwab, 2016). In generic terms, the twenty-first century features the informational interconnection between economies and the dynamics of globalization, leading the global economy towards increased sensitivity in turbulence (Vlados et al., 2018).

Furthermore, environmental history and the energy system’s structuration play a critical role in transport’s history since they are all complementary and mutually enriching fields (Bradford, 2018; Zeller, 2014). Nowadays, climate change, caused by energy-intensive activities, induces a paradigmatic change in transport, progressively cultivating a world without oil (Dimitriou, 2016; Munn, 2008). This revolution in transport creates conditions for innovation in this industry that could increase energy efficiency and reduce emissions—however, such a significant change necessitates large scale and establishment of new standards (Carlton et al., 2017; Hoppe & Winter, 2015). Moreover, in today’s era of globalization’s crisis, reinforcing local development seems capable of countering the consequences of climate change since it

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<sup>1</sup> The pre-capitalist evolution of the transport system is not in the scope of this study.

could mitigate transport-intensive imports—especially in goods such as food (Chrzanowski, 2019).

In an approach of significant clarity and usefulness, Vinichenko (2021) comprehensively addresses the problem of discovering critical trends in today's transport system. First, in cargo transportation (containerization), an increase in international trade will necessarily increase the demand for this mode since it is the most inexpensive and convenient. However, this opportunity is not equal for all carriers, who must invest in their competitive advantages and protect their business models from adjacent road network inefficiencies and low-performance indicators. Second, the use of artificial intelligence in freight transport will spread rapidly, applying methods such as “unmanned cars,” drones for “last mile” delivery, and “smart” infrastructure capable of reading urban objects and embedding them in the present infrastructure. Third, the greening of freight transport and de-carbonization will also cause significant transformations in the cargo flows and the composition of inventory flows. The use of biofuels, hydrogen energy, hydroelectric power, and the “battery-electric transmission” in the automobile industry will accelerate this transition. With exceptional clarity, Vinichenko (2021) also documents the tendencies occurring in passenger transport and tourism. First, the author notices that transports will continue to elevate as individualized tourist products—the customer's choice will rise as the primary success factor due to the gradual erasing of physical boundaries and the increased informatization, which allows individual tourist planning and optimal routes. Second, high-speed highways will further expand as passenger transport corridors, reassuring enhanced safety and access. Third, the electrification of private cars will significantly alter all transport infrastructure since the travelers will plot their leisure routes according to the charging stations' locations.

Overall, the trends forecasted about the global socio-economic system in the related scholarly literature seem to broadly correspond to the generalized socioeconomic depression that induces investment infrastructure and institutional innovations, together with reinforced digitalization and the diffusion of cyberphysical systems. Concerning the specific tendencies emerging in the transport industry, the literature increasingly suggests the electrification of automobiles (ideologically, the need for climate change protection and de-carbonization cultivates this change) and the expansion of artificial intelligence in demand-side transport solutions. The following section explores significant developments caused by the current pandemic, aiming to examine whether this crisis accelerates these trends.

### **3. COVID-19 as an accelerator of developments in the transport sector**

The previous section highlighted the critical trends that (as it seems) accelerate today's transition in the post-COVID-19 world, both in the global socioeconomic system and the

transport sector. According to the examined literature, the pillars concerning the global evolutionary transformation are the generalized socioeconomic depression, the investment in infrastructure, and the digital transformation. The two spheres specifically related to the transport industry are the transition to electric cars and de-carbonization and the enhancement of demand-side transport solutions.

Since the end of 2019, when some patients in Wuhan, China, showed atypical pneumonia symptoms, a profound and multilayered crisis started (both health crisis and socioeconomic crisis). Characteristically, the United Nations (2020) notice that the current pandemic crisis will probably lead to new famines and cause a backslide in poverty alleviation worldwide. This section investigates whether this crisis accelerates these five trends (see the earlier section) on the global socioeconomic system and the transport sector identified by the literature:

- A. Generalized socioeconomic depression: According to the International Monetary Fund (IMF, 2021), the world output recorded an overall fall of 3.3 percent, 4.7 percent in the advanced economies, and 2.2 percent in the emerging economies, with the world volume trade in goods and services declining by 8.5 percent in 2020. In a similar vein, the World Trade Organization suggests that this crisis will surpass in magnitude the financial crisis of 2008-2009 (Azevêdo, 2020). In the transport industry, a typical example of the recession is the global air traffic contraction by 53.3 percent in March 2020, according to the Committee for the Coordination of Statistical Activities (CCSA, 2020), and the ship call reduction of 8.7 percent in the first 24 weeks of 2020 compared to 2019 (UNCTAD, 2021). In terms of social implications, various analysts argue that human freedom and democracy are facing significant pressures since the pandemic started, pointing to a new “crisis of authority” and increasing inequalities and social injustice after the extended lockdowns across the world (European Parliamentary Research Service, 2020).
- B. Investment infrastructure inducement: According to the 10 United Nations Conference on Trade and Development (UNCTAD, 2020), investment reduction due to the pandemic has reached 30-40 percent worldwide. However, the research on the need for increased investment in infrastructure is growing over the past months (International Transport Forum, 2021b). According to the Global Infrastructure Hub (GIH, 2020), investment infrastructure should be the fundamental direction for long-term development in the G20 economies; this report bases its findings on the significantly bigger fiscal multiplier when infrastructure investment covers a substantial share of total public spending. Also, a considerable amount of research suggests the need for reinforced investment infrastructure in the post-COVID-19 transition, capable of favoring long-term balanced development across the planet (International Energy Agency, 2020a; International Transport Forum, 2021a; Tandberg & Allen, 2020).

- C. Digitalization: According to the International Labor Organization (ILO, 2020), only 18 percent of the labor force globally works today in an environment capable of offering infrastructure for remote working. However, the increased digitalization that happened since the pandemic is noteworthy. According to the Organization for Economic Co-operation and Development (OECD, 2020), e-commerce sales increased by 30 percent in April 2020 compared to 2019 in the EU (retail sales contracted in the same period by 17.9 percent). Also, the joint research by UNESCO, UNICEF, and the World Bank (UNESCO et al., 2020) shows that most countries worldwide promoted at least one policy to support distance learning. At the microeconomic level, a survey by the World Economic Forum (2020) in various business leaders about the future of jobs concludes that 80 percent of the respondents have stimulated in-business digitalization (among other things, 80 percent declared remote work as the primary move and 50 percent automation). Overall, a growing number of scholars examine today's acceleration of digital transformation due to the pandemic (Soto-Acosta, 2020; Winarsih et al., 2021).
- D. De-carbonization and electrification of automobiles: According to the International Energy Agency (2020b), energy demand, energy investment, and energy-related CO<sub>2</sub> emissions will record a significant decline due to the pandemic (energy demand fell by 5 percent in the first quarter of 2020). The worldwide lockdown measures contributed to this contraction; however, policymakers show an increasing will to keep up with these trends since de-carbonization goals seem achievable now, such as the climate-neutral EU by 2050 (Karlsson & Silander, 2020). Concerning electric mobility, the COVID-19 crisis led car registrations globally to a significant reduction, with the share of electric vehicles upsurging. As noticed by a relevant report of the International Transport Forum (2020), automobile production and sales recorded an 80 percent decline in China in February 2020 compared to 2019 and 50 percent in March, respectively. Also, this fall reached 85 percent during March in Italy, more than 70 percent in France, and 70 percent in Spain (a similar trend appeared in other developed economies also). At the same time, electric vehicles sales tripled in the United Kingdom in March 2020 compared to March 2019, while Italy recorded a 50 percent increase and France 20 percent.
- E. Demand-side reinforcement in transport solutions: The COVID-19 crisis led to the further development of "prosumerism" because it benefits organizations to stay resilient in today's turbulent socioeconomic environment. The "prosumer" is the individual who consumes and produces value simultaneously (for consumption by others or self-consumption), receiving explicit or implicit incentives from the organizations involved in the exchange (Lang et al., 2020). Since the advent of the pandemic, the sharing economy experienced significant growth, offering new service delivery solutions by substituting traditional delivery methods (for instance, see the following: Business

Insider, 2020; O'Brien, 2020; Reuters, 2020; UBER, 2021). Also, according to the International Transport Forum (2020b), drones' use increased remarkably during the pandemic because drones helped transport supplies with minimum human interaction and protection from the virus. This fundamental transformation in transport demand because of the crisis will lead to substantial changes in public transport and, as reported by the International Transport Forum (2021c), different transport services will emerge in the private and the public sphere. New applications that thoroughly examine passenger volume and mobility patterns will arise in public transport, necessitated by health concerns. According to Quiros (2020), various private companies already use big data and digital technology to assist public transport organizations design agile and responsive routes to meet changes in transport patterns.

Overall, all trends point to an explicit acceleration nowadays. The global socio-economic system enters a paradigmatic transition (in the techno-economic sphere) without knowing when the actual metamorphosis will occur. The fast-paced digitalization, the need for infrastructure investment, and the structural change of the energy sector highlight the emerging context of the post-COVID-19 era and fourth industrial revolution. The transport industry, which was hit hard by the pandemic, is expected to continue changing towards the directions deriving primarily from the de-carbonization diffusion and the reinforcement of the consumer (the "prosumer").

#### **4. Final remarks and prospects**

This presentation ventured on an elliptical review of today's transport sector. The goal was to prove its transition to the post-COVID-19 era. The contribution of this introduction is the reaffirmed acceleration taking place nowadays due to the pandemic in most fields of the global socioeconomic system and, specifically, in the transport industry.

In the future, this research could examine more implications caused by this crisis to globalization and enrich the debate. Related research is already underway, the findings of which highlight the need to reposition the perception of global governance in the post-COVID world (Vlados & Chatzinikolaou, 2021). Also, the historical analysis of other industries could help strengthen this "acceleration argument" in a future setting. Finally, this repositioned theoretical framework for the transport sector evolution could lead to action research in specific transport companies (in specific spatial contexts) to find how they respond and adapt in light of these findings.

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